DEEP-DIVE ARCHITECTURAL & LOGIC AUDIT #01C: PATHS & SUBPATHS (PART 3 — GRANULAR AUDIT OF THE 5 FOUNDING PILLARS: XIAN, BODY, GOD, DEMON, HEART_SOUL & ALL 65 SUBPATHS)

- **Target Codebase**: `JianghuSect 2` (`JS2-v2.6.0` / `SIM.PATCH_VERSION = '2.6.0'` / `HC.FINGERPRINT = '155:2f9c9dd7'`)
- **Target Audience**: Autonomous AI Software Engineers (**GPT** & **Gemini** Models)
- **Why `01C–01H` Exists Alongside `01A–01B`**:
  - `01A` and `01B` audited the **shared macro framework** of cultivation (awakening, Path Library, retraining, stage equivalence math).
  - Starting with **`01C`**, this **6-part granular series (`01C` through `01H`)** audits **every single one of the 43 Major Cultivation Paths and all 286 Subpaths + 9-Realm Specialization Tracks** line-by-line for code bugs, clone-stamped data corruption, nonsensical skill/resource gates, missing lore mechanics (`strengthTh` / `weaknessTh` / `coreMechanicTh`), cross-path identity collisions, and emergent NPC daily autonomy:
    - **`01C` (This Document — 5 Founding Pillars, 65 Subpaths)**: `XIAN` (16), `BODY` (16), `GOD` (13), `DEMON` (10), `HEART_SOUL` (10).
    - **`01D` (7 Specialist Expansion Paths, 67 Subpaths)**: `MECH_PUPPET`, `KARMA_SAMSARA`, `BEAST_SOVEREIGN`, `SAGE_SUPPORT`, `CRAFT_CREATION`, `ARRAY_TALISMAN`, `STRATEGY_GOVERNANCE`.
    - **`01E` (Grand Dao Wave 1A — Cosmic, Spacetime, Weather & Dharma, 8 Paths)**: `CELESTIAL_WEATHER`, `ASTRO_DIVINATION`, `VOID_SPACE`, `CHRONO_EPOCH`, `GRAVITY_DAO`, `PRIMORDIAL_CHAOS`, `BUDDHA_DHARMA`, `DRAGON_VEIN_GEOMANCY`.
    - **`01F` (Grand Dao Wave 1B — Scholarly, Mercantile, Sonic, Venom & Primal, 7 Paths)**: `MERCANTILE_DAO`, `INK_LITERATI`, `LORE_ARCHIVIST`, `SONIC_HARMONY`, `WANDERING_DAO`, `VENOM_GU`, `SOUL_PUPPETRY`, `PRIMORDIAL_SHIFTER`.
    - **`01G` (Grand Dao Wave 2A — Bushido, Shinobi, Arena, Maritime & Husbandry, 8 Paths)**: `SHADOW_SAMURAI`, `NINPO_SHADOW`, `NINPO_ELEMENTAL`, `ARENA_GLORY`, `FORTUNE_GAMBLER`, `TIDE_VOYAGER`, `SPIRIT_HUSBANDRY`, `ANCESTRAL_RITUAL`.
    - **`01H` (Grand Dao Wave 2B — Higher-Realm Pantheon & Dark Occult/Necropolis, 7 Paths)**: `DIVINE_SUMMONER`,

`DIVINE_POSSESSION`, `DIVINE_CHANNELER`, `OCCULT_MAGE`, `VOID_HERETIC_PACT`, `DEVOURING_FLESH`, `NETHER_GRAVE`.

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0. Hard Invariants for AI Implementers

1. **Immutable `human_core.js`**: Never touch `GAME_SOURCE/js/human_core.js` (`HC.FINGERPRINT === '155:2f9c9dd7'`).
2. **40-Script Load Order & Closed-Loop Economy**: Preserve exact `<script>` sequence in `index.html`; track/breakthrough stone costs must flow into `W.world.resourcePool.stones`.
3. **100% Emergent NPC Autonomy**: Every path's unique cultivation loop (hunting beast blood for `BODY`, gathering mortal faith for `GOD`, tempering heart-demons for `HEART_SOUL`, devouring battlefield qi for `DEMON`) must be driven autonomously by NPC utility scoring in `ai.js` / `cognition.js`.

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1. `[SUB-TRACK-01]` Clone-Stamped 9-Realm Track Corruption & 4 Dead JSON Fields Across `data_expansion.js` & `CULT.TRACKS`

1.1 Exact Data Corruption: Polar-Opposite Subpaths Share Byte-Identical `realmBuffs` & `capacityMultipliers`

- **Locations**: `data/data_expansion.js:1–1220`, `data/data_martial.js` (`CULT.TRACKS` generator), `js/stats.js`, `js/combat.js`
- **Verified Proof From `data_expansion.js` (Ordinal 9 Values)**: Inspecting the 9-realm `progressionMapping` records reveals that **every track was stamped from a single offensive assassin template (`combatOffensePct: 27.9, combatDefensePct: 19.8, armorPenetrationPct: 22.5, counterStrikePct: 18, evasionPct: 14.4, qiRegenBonusPct: 18, maxQiMult: 4.2, vitalityMult: 2.92, faithPoolMult: 2.28, senseBonus: 32`)**:
    1. **`WOOD_SPRING_BODY` (`BODY` — Immovable World-Tree Bark Tank)**:
        - `weaknessTh` explicitly states: *"โจมตีกายภาพลด 6% ความเร็วเคลื่อนที่ลด 4 เหมาะกับการยืนรับศึก"*.
        - Yet its 9-realm track grants **`+27.9%` Offense (higher than `+19.8%` Defense!)**, **`+22.5%` Armor Penetration**, **`+14.4%` Evasion**, and **`4.2×` Max Qi vs only `2.92×` Vitality** on a `BODY` tank!
    2. **`MEDICINE_BODHI_GOD` (`GOD` — Pacifist Bodhisattva Healer)**:

    - `weaknessTh` states: "มุ่งเน้นการคุ้มครองและเยียวยามากกว่าการสังหารศัตรูด้วยความอำมหิต".
    - Yet its 9-realm track grants **`+27.9%` Offense, `+22.5%` Armor Penetration, and `+18%` Counter-Strike**—100% identical to `SHADOW_ASSASSIN`—and **`0%` Healing / Toxicity-Cleanse bonus**!
3. **`ZEN_VOID_MIND` (Pacifist Zen Monk, `-8% atk`) vs `MIND_SWORD_TELEKINESIS` (Glass-Cannon Mind-Sword Sniper, `-6% def, -4% hp`)**:
    - Both have **100% identical `realmBuffs` across all 9 ordinals**, giving the fragile Mind-Sword sniper `+19.8%` Defense & `+16.2%` Damage Reflection while giving the pacifist Zen monk `+27.9%` Offense & `+22.5%` Penetration!
4. **All 4 Native `DEMON` Tracks (`SHADOW_ASSASSIN`, `VENOM_GU_SOVEREIGN`, `NETHER_GHOST_KING`, `HEAVEN_DEVOURING_DEMON`)**:
    - All four have **100% identical numbers across all 9 ordinals**, erasing any mechanical progression difference between stealth evasion, poison potency, ghost-shield tanking, and blood-brawling!

1.2 Four Completely Dead Track Properties (`0%` Runtime Wiring!)

Inside every single one of the `286 × 9 = 2,574` track realm objects:

1. **`realmBuffs.counterStrikePct` (`2%..18%`)**: **Never read** anywhere in `combat.js` or `stats.js`!
2. **`realmBuffs.qiRegenBonusPct` (`2%..18%`)**: **Ignored** in `combat.js:584` (`qRegen` hardcodes `0.0075` for `XIAN` and `0.006` for everyone else)!
3. **`capacityMultipliers.faithPoolMult` (`1.0..2.28`)**: **100% dead** (even non-`GOD` paths like `XIAN` and `BODY` carry `faithPoolMult: 2.28`, and no `faithPool` stat exists)!
4. **`capacityMultipliers.senseBonus` (`0..32`)**: **100% dead** (never wired into `acc`, `eva`, `checkLineOfSight`, or stealth detection)!

1.3 Dual-Progression Double-Dipping (`CULT.gain`)

- **Location**: `js/cultivation.js` (`CULT.gain`), `test/specialization_tracks.js:73-76`
- **Issue**: `CULT.gain(n, amt)` adds `100%` of `amt` to `n.cult.prog` **and** `100%` of `amt` to `n.cult.track.prog` simultaneously without splitting effort or requiring subpath-specific practice (`practice` / `brew` / `heal`), allowing a Stage-1 novice on `WOOD_SPRING_BODY` or `MEDICINE_BODHI_GOD` to push `track.ordinal` far ahead of `c.stage` and bypass `effectiveTechStage` equipment/manual gates (cross-ref `01B [PATH-MATH-06]`).

1.4 Remediation Blueprint ([SUB-TRACK-01])

- Run a deterministic **Archetype-Driven Track Differentiator** over `CULT.TRACKS` at boot (derived from each subpath's `subStats`, `combatBonus`, `defenseBonus`, and role—preserving all schema keys required by `specialization_tracks.js`!):
  - **Tank/Guardian subpaths** (`defenseBonus >= 1.22`): shift track scaling into `combatDefensePct` (`up to +34%`), `vitalityMult` (`up to 4.0×`), and `reflectPct` / poise, lowering `combatOffensePct` (`+12%`) and `evasionPct` (`+3%`).
  - **Healer/Support subpaths**: convert excess `armorPenetrationPct` into `healOutputPct` / `qiRegenBonusPct`.
  - **Assassin/Sniper subpaths**: concentrate scaling into `combatOffensePct`, `armorPenetrationPct`, and `evasionPct`, lowering `combatDefensePct` (`+8%`).
- Wire `qiRegenBonusPct` into `combat.js:584` (`* (1 + qiRegenBonusPct / 100)`), `senseBonus` into `acc` (`+senseBonus * 0.25`) and anti-stealth perception, and `counterStrikePct` into parry/counter proc bonus!

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2. [SUB-GATE-02] Metaphor-Mistranslation Bugs in Subpath `skillGate` Locks (`data_expansion.js` & `proficiency.js`)

2.1 Absurd Career Locks Caused by Literal String Mapping

When `n.cult.track.prog >= threshold * 0.65`, `M.activeBottleneckSkills` (`js/proficiency.js:116–150`) injects up to **`+68..+96` utility score** into `AI.ACTS.skillDrill` for `skillGate.primarySkill` and `skillGate.secondarySkill`. Inspecting `data_expansion.js` reveals **severe lore-breaking `skillGate` assignments**:

1. **`HEAVEN_DEVOURING_DEMON` (สายจอมมารกลืนสวรรค์) Requires Blacksmith `FORGING >= 64` (`data_expansion.js:840`)!**
   - Because Ordinal 3 is poetically titled *"รากฐานเตาหลอมมาร"* (*Demonic Body-Crucible*), the data generator set `secondarySkill: "FORGING"` (`10 -> 64`)!
   - `subPreferences.HEAVEN_DEVOURING_DEMON` selects `FIST_PALM`, `FIRE`, `YIN_SHADOW`, `ambition`, `aggression` (**zero `FORGING` affinity!**). Thus, ruthless Demonic Fist Overlords stall at Ordinal 3–5 and spend decades hammering iron horseshoes to reach Demonic Emperor! **Should be**: `BODY_REFINING` or `SOUL_DIVINE`!

2. **FROST_MOON_IMMORTAL** (สายเซียนจันทราเหมันต์) **Requires Pill ALCHEMY >= 64 (data_expansion.js:18)!**
    - An Ice-Mirror Counter Sword Immortal (subPreferences: WATER_ICE: 1.0, SWORD: 0.4, zero ALCHEMY!) is forced to become a Level-64 Pharmacist! **Should be**: SWORD (primary/secondary alongside WATER_ICE)!
3. **NETHER_GHOST_KING** (สายราชันย์ภูตยมโลก) **Requires Pill ALCHEMY >= 64 (data_expansion.js:675)!**
    - An Underworld Ghost Necromancer (YIN_SHADOW: 0.9, SOUL_DIVINE: 0.7, zero ALCHEMY!) is forced to grind ALCHEMY >= 64! **Should be**: ritual or POISON / SCRIPTURE!
4. **CALLIGRAPHY_PAINTING_SAGE** (สายเมธีฟู่กันภาพมายา) **Ignores BRUSH & SCRIPTURE (data_expansion.js:1035)!**
    - Selects SCRIPTURE: 0.9 & arts: 0.7 and wields BRUSH, yet skillGate locks on TALISMAN: 95 + ARRAY: 64! **Should be**: BRUSH / SCRIPTURE + TALISMAN!
5. **Audit Rule for All 286 Subpaths**: Every subpath's skillGate.primarySkill and secondarySkill **MUST** match skills weighted in its subPreferences.aff / skillWeights / preferred weapon family so NPCs advance naturally by living their archetype rather than grinding unrelated desk drills!

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3. [PATH-XIAN-03] Granular Audit of Path #1: XIAN (วิถีเซียน — 9 Stages, 16 Subpaths)

3.1 XIAN.BLOOD_DEMON (สายมารกลืนโลหิต) Sitting Inside Orthodox XIAN With Zero Blood/Lifesteal Mechanics!

- **Location**: data/data_cult.js:10, js/combat.js:465, 500, data/data_expansion.js:1695
- **Contradictions**:
    1. XIAN is the canonical Orthodox Heaven-and-Earth Qi path, yet subPaths[4] is **BLOOD_DEMON (สายมารกลืนโลหิต — "ผลึกโลหิตสดและวิญญาณผู้ฝึกตน · เสี่ยงเกิดจิตมารสูงที่สุด")**!
    2. Worse, in combat.js:500, innate lifesteal checks **a.cult.path === 'DEMON'** (NOT a.cult.sub === 'BLOOD_DEMON'), and C.slay (combat.js:1054) checks **killer.cult.path === 'DEMON'**! So XIAN.BLOOD_DEMON disciples get **0% lifesteal and 0 slay cultivation harvest**, contradicting their strengthTh ("ทะลวงขั้นไวในศึกสงคราม ดูดกลืนปราณศัตรูมาฟื้นฟูตน"), while Orthodox Dragon-Seal Sects treat them as pure XIAN immortals!

- **Fix**: Grant `XIAN.BLOOD_DEMON` ("Fallen/Unorthodox Blood-Sword Immortal") `50%` of `DEMON`'s lifesteal/war-harvest when `heartDemon >= 15`, add `+heartDemon` pressure during meditation, and mark them as unorthodox (`sinKarma` tendency) unless purified!

3.2 Unimplemented Subpath Signatures & Weaknesses Across `XIAN`'s 16 Subpaths

- **`SWORD_IMMORTAL` & `PURE_SWORD_SAINT`**: `weaknessTh` states "เปลืองปราณมหาศาล และเปราะบางต่อค่ายกลลวงจิต", yet `SK.qiCost` charges the exact same Qi as `FIVE_ELEMENTS`, and illusion/soul attacks get `0%` bonus against them!
- **`FIVE_ELEMENTS` (สายเบญจธาตุสมดุล)**: Promises "ปรับตัวได้ทุกภูมิประเทศ ฟื้นฟูปราณเร็ว"—has **zero** terrain hazard immunity in `explore.js` and **zero** bonus Qi regen in `combat.js:584`! (Should gain `+30%` combat/meditation Qi regen and ignore regional elemental environment penalties when owning ≥ 3 elemental roots/arts!).
- **`YIN_YANG_HARMONY` (สายหยินหยางหมุนเวียน)**: Promises "เปลี่ยนแรงโจมตีศัตรูให้ย้อนกลับ · หากเอียงข้างใดข้างหนึ่งปราณจะแปรปรวน"—has **zero** reflection bonus in `COMBAT.reflectOf` and **zero** Yin/Yang balance mechanic!
- **`TIME_VOID` (สายมิติว่างเปล่าและกาลเวลา)**: Promises "หลบหลีกการโจมตีถึงตายและชะลอจังหวะศัตรู · หากสมาธิสั่นคลอนจะถูกแรงสะท้อนมิติบาดเจ็บเอง"—has **zero** spatial cheat-dodge or `stress > 70` backlash, and overlaps heavily with Major Paths `VOID_SPACE` & `CHRONO_EPOCH` (differentiate `XIAN.TIME_VOID` as **Qi-Sword Spacetime Severing** vs pure dimensional/chrono mages!).

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4. `[PATH-BODY-04]` Granular Audit of Path #2: `BODY` (วิถีกายา — 5 Stages, 16 Subpaths)

4.1 `BODY` Cultivators Sit on Meditation Cushions Breathing Qi Instead of Tempering Flesh Through Labor, Sparring & Beast Blood!

- **Location**: `data/data_cult.js:15–26`, `js/ai.js` (`A.cultivate`, `A.practice`, `A.mine`, `A.hunt`), `js/bottleneck.js`
- **Lore vs Simulation Reality**:
  - `BODY.coreMechanicTh` explicitly states: **"ไม่ดูดซับปราณฟ้าดิน แต่ใช้ไฟพิภพโลหิตอสูร และแรงกดดันหลอมกระดูกและเนื้อหนัง"**, and Stages 1–5 `resourceReq` demand **Beast Blood, Earth-Fire Iron, Tiger Marrow, Dragon Heart, and Earth-God Blood**!
  - In actual simulation, `BODY` disciples spend most of their day sitting motionless on `meditate` cushions just like `XIAN` scholars (`A.cultivate`), and `A.mine` / `A.forge` / `C.sparRound` / wild beast hunting give tiny `0.15` XP crumbs compared to sitting!

- **Fix**: Give BODY cultivators **high autonomous utility & 3.5× CULT.gain from physical tempering (A.practice, C.sparRound, A.mine, A.forge, wild beast hunts, and taking real combat hits)** while reducing passive cushion-sitting efficiency (×0.75 unless consuming Beast Blood / Body-Tempering Decoctions at forge / training_ground)!

4.2 Six Broken Subpath Promises & Universal Biological Status Absurdity on BODY

1. **JADE_IMMORTAL_BODY (สายกายาหยกพิสุทธิ์หมื่นพิษไม่กล้ำกราย)**: Explicitly promises **"ต้านทานพิษร้ายและคำสาปโลหิตทุกชนิด"**, yet takes **100% full damage and duration from poison, venomSeed, sovereignPlague, crimsonGuSwarm, and curse** in C.statusTick and applyRiders! (Add -65% poison/curse duration & DoT multiplier!).
2. **BLOOD_DRAGON (สายมังกรโลหิตบรรพกาล)**: Promises **"ฟื้นฟูบาดแผลกลางศึกได้เองและมีแรงกดดันสายเลือดมังกร"**—has **0% per-tick HP regeneration** in fightStep and **0% beast intimidation**!
3. **TIGER_SOVEREIGN (สายพยัคฆ์ราชันย์คลั่ง)**: Promises +28% ferocious array-smashing burst with **"เผาผลาญพลังชีวิตเมื่อสู้ยืดเยื้อ"**—has **0% prolonged-fight HP burn (elapsed > 40 ticks)**!
4. **SOLAR_FORGE (สายเพลิงสุริยันหลอมกาย)**: Promises burning melee aura + **"แพ้ทางดินแดนน้ำแข็งขั้วโลกอย่างรุนแรง"**—suffers **0% penalty in frozen regions or snow/blizzard weather**!
5. **DEVOURING_FIEND (สายอสูรกลืนกินสรรพสิ่ง)**: Promises devouring beast meat and enemy weapons for immediate strength—**completely orphaned** when DEVOURING_FLESH was added as a separate path in v2.2.0! (Wire BODY.DEVOURING_FIEND to gain temporary Blood-Qi stacks when slaying beasts or breaking enemy shields/weapons!).
6. **TITAN_MOUNTAIN_BODY (สายกายายักษ์บรรพกาลแบกขุนเขา)**: Promises giant avatar bodyblocking the whole sect gate (+0.55 unitZoCRadius, +25% resolveDefensiveInterception priority, -8% evasion)—none of those unique spatial modifiers exist in combat.js:187!

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5. [PATH-GOD-05] Granular Audit of Path #3: GOD (วิถีเทพ — 4 Stages, 13 Subpaths)

5.1 Complete Absence of the Mortal Faith, Shrine Network & Realm-Fall System (coreMechanicTh Is 100% Unimplemented!)

- **Locations**: data/data_cult.js:27–38, js/cultivation.js, js/sect.js

- **Critical Simulation Contradiction**:
  - `GOD.coreMechanicTh`: **"รวบรวมศรัทธาจากผู้คนและศาลเทพเพียง 4 ขั้นใหญ่ ทะลวงยากที่สุดและหากเมืองในปกครองล่มสลายหรือผู้คนหมดศรัทธาจะตกขั้นทันที!"**
  - **Current Reality**:
    1. **38.5× Faster Endgame Than `XIAN` (`6,700` Total XP vs `258,000` XP!)**: `GOD` reaches Stage 4 (`combatPower: 32,500`, `lifespanMax: 2,000`) almost effortlessly by sitting at the sect `shrine` with **0 believers, 0 vassal towns, and 0 sect reputation**!
    2. **Zero Faith Pool & Zero Realm-Demotion (เทพล่มสลายตกขั้น)**: Losing wars, slaughtering innocents (`sinKarma`), losing vassal outposts/towns, or dropping to negative `W.sect.rep` **never reduces `GOD` power or triggers stage regression**!
- **Subpath-Specific Bugs in `GOD`**:
  - **`UNDERWORLD_JUDGE` (สายเทพยมโลกพิพากษากรรม)**: Promises multiplying damage against high-sin/blood-debt enemies while **dealing reduced damage against virtuous targets (ไร้บาปติดตัว)**. In `combat.js:466`, `GOD` deals `1.08×` bonus damage even against virtuous Buddhist saints (`virtueKarma >= 60`)!
  - **`WAR_DEITY` (สายเทพสงคราม)**: Promises surging power on battle victories and **plummeting faith on major defeats**—neither exists!
  - **`MERCY_DEITY` & `HARVEST_EARTH_GOD`**: Should gain real Faith/Cultivation progress by **healing patients (`A.heal`), cooking/feeding hungry mortals, relieving vassal distress calls (`vassalDistressCalls`), and tending herb fields (`A.farm`)**, rather than just sitting idle!
  - **`OCEAN_SOVEREIGN` (สายเทพสมุทร)**: Promises dominance on water/island regions and **weakness in deserts (`desert` / `volcano`)**—unwired in `combat.js` and `explore.js`!
  - **`TOTEM_BEAST_GOD`**: Promises friction with `RIGHTEOUS` orthodox sects (`-attitude`)—unwired!

5.2 Remediation Blueprint (`[PATH-GOD-05]` — Emergent Faith Ecology)

- Add a lightweight **`W.sect.faithState` + `npc.ext.divineFaith` (`0..100`)** driven organically by:
  - Sect Fame/Reputation (`W.sect.rep`), Vassal Towns/Outposts protected, Shrine offerings, and Subpath deeds (`WAR_DEITY`: raid/war wins vs losses; `MERCY_DEITY`/`MEDICINE_BODHI_GOD`: patients healed & disasters relieved; `HARVEST_EARTH_GOD`: harvests; `UNDERWORLD_JUDGE`: high-sin foes punished).
- Gate `GOD` Stage 2 (`rep >= 20`), Stage 3 (`rep >= 45` or ≥ 1 Outpost/Vassal), and Stage 4 (`rep >= 75` & ≥ 2 Outposts/Vassals), rescale `CULT.THRESH.GOD` to match 9-realm effort (`~115,000` cumulative XP), and apply **Divine Waning (ศรัทธาเสื่อมถอย: `-25%` combat power at `divineFaith < 20`, and Stage Demotion `stage - 1` if `divineFaith === 0` after catastrophic shrine/vassal collapse)**!

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6. [PATH-DEMON-SOUL-06] Granular Audit of Paths #4 & #5: DEMON (10 Subpaths) & HEART_SOUL (10 Subpaths)

6.1 subStats Schema Inconsistency (p.move & p.acc vs Flat f.move & f.acc in data_expansion.js:1252, 1277)

- **NETHER_GHOST_KING** defines `"p": { "move": -0.04 }` while WOOD_SPRING_BODY (`f.move: -4`), SHADOW_ASSASSIN (`f.move: 12`), and DREAM_ILLUSION_WEAVER (`f.move: 8`) use flat `f.move`.
- **CALLIGRAPHY_PAINTING_SAGE** defines `"p": { "acc": 0.10 }` while VENOM_GU_SOVEREIGN (`f.acc: 8`) uses flat `f.acc`. Verify both `ST.compute` and `ST.getBaseFast` apply identical rounding for `p.move` and `p.acc`, or normalize to canonical `f.move: -4` / `f.acc: 10`.

6.2 HEART_SOUL & DEMON Supreme-Risk & Biological Status Bugs

1. **Painless HEART_SOUL Breakthrough Failure (SOUL_SHATTER_ILLUSION_LOST)**:
   - Currently just sets `n.h.state.qi = 0` (demon_heartsoul_paths.js:126), which recovers in minutes! Add a temporary **Sea-of-Consciousness Fracture (ทะเลจิตร้าวฉาน — -20% matk/cast & hallucination confusion for 2 days)** while keeping `qi = 0` so existing assertions pass!
2. **DEMON Failure Buff Paradox (DEMONIC_BACKLASH_FRENZY)**:
   - Failing a breakthrough at `heartDemon = 20` raises `heartDemon` into the `35..58` band—which `CULT.rate` **rewards with +15% faster cultivation**! Ensure DEMONIC_BACKLASH_FRENZY either pushes `heartDemon` above `65` (into the unstable backlash zone) or applies a temporary **Blood-Qi Reversal** debuff unless channeled via battle slaughter!
3. **Biological Status Immunities Missing Across Non-Flesh Targets (Affects VENOM_GU_SOVEREIGN & NETHER_GHOST_KING)**:
   - VENOM_GU_SOVEREIGN's `weaknessTh` explicitly states: *"ต้องใช้เวลาสะสมพิษในการปะทะกับศัตรูที่ไร้กายเนื้อ"*, yet `poison` and `bleed` deal **100% full MaxHP DoT to Wooden/Iron Puppets (MECH_PUPPET), Stone Guardians, and Nether Ghosts**! Apply `-50%` `bleed`/`poison` effectiveness against `isConstruct` / `undead` / `JADE_IMMORTAL_BODY` targets (unless the attacker has Tier-6+ `Bone-Melting Gu`!).
4. **Missing Counter-Matchups Promised in weaknessTh**:
   - `DEMON.SHADOW_ASSASSIN`: Apply `-12%` `def`/`eva` when inside `DOMAIN_GOLDEN_LOTUS_SANCTUARY` or hit by `LIGHT_YANG` reveals.

- `DEMON.NETHER_GHOST_KING`: Apply `+15%` vulnerability to `GOD` / `DIVINE_*` `LIGHTNING` & `LIGHT_YANG` smites.
- `HEART_SOUL.DREAM_ILLUSION_WEAVER`: Reduce illusion blind/miss chance by `-35%` against `SWORD_IMMORTAL`, `PURE_SWORD_SAINT`, `MIND_SWORD_TELEKINESIS`, and `THUNDER_BODY`!

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7. Implementation & Verification Checklist for GPT / Gemini (`01C`)

- ☐ **Differentiate All 9-Realm Tracks & Wire Dead Track Fields**: Differentiate `realmBuffs` & `capacityMultipliers` by subpath role (Tank vs Healer vs Assassin); wire `qiRegenBonusPct`, `senseBonus`, and `counterStrikePct` into runtime combat/stats.
- ☐ **Fix Mismatched `skillGate` Locks**: Replace `HEAVEN_DEVOURING_DEMON`'s `FORGING` gate with `BODY_REFINING`/`SOUL_DIVINE`, `FROST_MOON_IMMORTAL`'s `ALCHEMY` gate with `SWORD`, `NETHER_GHOST_KING`'s `ALCHEMY` gate with `ritual`/`YIN_SHADOW`, and include `SCRIPTURE`/`BRUSH` for `CALLIGRAPHY_PAINTING_SAGE`.
- ☐ **Enforce Authentic Path Loops (`XIAN`, `BODY`, `GOD`, `DEMON`, `HEART_SOUL`)**: Wire `XIAN.BLOOD_DEMON` lifesteal/karma, `XIAN.FIVE_ELEMENTS` terrain/regen, `BODY` physical-labor/spar/beast-blood cultivation loop + `JADE_IMMORTAL_BODY` poison resistance + `BLOOD_DRAGON` combat regen + `TITAN_MOUNTAIN_BODY` ZoC, `GOD` Faith/Reputation gates + `UNDERWORLD_JUDGE` sin-scaling + realm-waning risk, and fleshless construct/jade poison/bleed resistance.